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Explain functional dependency and its types using relevant examples.

In a DBMS context, Functional Dependency (FD) is constraint to express the relationship of attributes (columns) of a table, how the value of one attribute uniquely defines the value of another.

The statement "Attribute B is functionally dependent on Attribute A" would mean that if we know the value of A, we can uniquely determine one value for B.

Notation: $X \longrightarrow Y$

- **X (Determinant):** The left side. The attribute that identifies the other.
- **Y (Dependent):** The right side. The attribute that is determined by X.

let's imagine our database table looks like this:

Student_ID	Student_Name	Project_ID	Project_Name	Instructor_ID	Instructor_Name
S01	David	P100	AI Chatbot	I-50	Dr. Bala
S01	David	P200	Web Portal	I-60	Prof. Ada
S02	Sarah	P100	AI Chatbot	I-50	Dr. Bala

Types of Functional Dependency

1. Trivial Functional Dependency

A dependency is trivial if the information on the right side is also present on the left side. Such dependency gives us no new information.

Definition: $X \longrightarrow Y$ is trivial if Y is a subset of X .

- Example: $\{Student_ID, Student_Name\} \longrightarrow Student_ID$

The problem I've identified is this: if I then tell you, "The student is S01, David" and prompt you with "What is the Student ID?" you already have the answer as I've just told you, and therefore don't need to consult the database to answer this.

2. Non-Trivial Functional Dependency

This is the valuable one. It applies when right side attribute is not an attribute on the left side. That will provide us with a new information.

Definition: $X \longrightarrow Y$ is non-trivial if Y is NOT a subset of X .

- Example: $Student_ID \longrightarrow Student_Name$

If I provide you with S01, you may access the table and conclude that the name is "David." "David" is not within "S01," and thus this is a valid, meaningful relationship.

3. Full Functional Dependency

This is what we are aiming for in database design. The state where one attribute depends on the full Primary Key.

Scenario: Let's say the Primary Key for our table is the combination $\{Student_ID, Project_ID\}$.

- Example: $\{Student_ID, Project_ID\} \longrightarrow Grade$ (Assume we add a Grade column).

Explanation:

- Is grade only determined by $Student_ID$ (S01)? No. David may receive grade A for AI Chatbot but B for Web portal project.
- Is grade only determined by $Project_ID$ (P100)? No. Since many students are involved in the AI Chatbot project, thus each student will receive a different grade.

Conclusion: To look for the grade, the combination of StudentID and ProjectID (S01 and P100) are both needed; therefore, the grade is totally determined by the entire key.

4. Partial Functional Dependency

This is a bad dependency as it creates redundancy (repetition) of the data. This occurs when a non-key attribute is only partially dependent on the Primary Key.

Scenario: Again, our Primary Key is {Student_ID, Project_ID}.

- Example: {Student_ID, Project_ID} $\longrightarrow$ Student_Name

Explanation:

- In order to retrieve the name David, must we know which project he is doing? No.
- We can retrieve the name David knowing only the Student_ID, S01.
- Since StudentName depends on StudentID alone, Student_ID being part of the primary key only, this is a Partial Dependency.

Why it's bad: If David joins 10 different projects, we have to repeat David 10 times in the database. If he changes his name, we have to update it in 10 places.

5. Transitive Functional Dependency

This occurs when there is an indirect relationship between attributes. It usually happens when a non-key attribute determines another non-key attribute.

Structure: A $\longrightarrow$ B and B $\longrightarrow$ C, therefore A $\longrightarrow$ C.

- Example:
 1. Project_ID $\longrightarrow$ Instructor_ID (P100 is always supervised by I-50).
 2. Instructor_ID $\longrightarrow$ Instructor_Name (I-50 is always Dr. Bala).
 3. Therefore: Project_ID $\longrightarrow$ Instructor_Name.
- Explanation:
 - The Instructor_Name depends on Instructor_ID.
 - The Instructor_ID depends on Project_ID.
 - So, Project_ID determines Instructor_Name *transitively* (indirectly).

Why it's bad: If Dr. Bala leaves and we delete his record from the Instructor table, but we still have him listed next to a project in this table, we create inconsistencies.

6. Multivalued Dependency (MVD)

Here is another, slightly more complicated, type; one attribute uniquely specifies a set of values of a second attribute. The values are not dependent on any other attribute in the table.

- Notation: X $\twoheadrightarrow$ Y
- Example: Student_ID $\twoheadrightarrow$ Phone_Numbers

Explanation:

- A student S01 (David) may have more than one number: '080123...' or '080987...'.

- If we try to enter this in one row: we face an issue; we may put more than one value in one cell (not desirable) or we might enter the Student's name twice and have 2 rows only to enter his numbers.
- This indicates that Student_ID is linked to a set of numbers rather than one value.

References

1. GeeksforGeeks

This article covers the 6 main types (Trivial, Non-Trivial, Multivalued, Transitive, etc.) with specific Roll No and Student examples.

[Types of Functional Dependencies in DBMS - GeeksforGeeks](#)

2. Guru99

This tutorial is the source for the `Student_Project` table structure (`Student_ID`, `Project_ID`, etc.) used to explain Transitive and Partial dependencies clearly.

[Functional Dependency in DBMS: Rules, Types & Examples - Guru99](#)

3. TutorialsPoint

This guide distinguishes clearly between Trivial and Non-Trivial dependencies using simple set notation examples

[Functional Dependency in DBMS - TutorialsPoint](#)

4. Simplilearn

This article connects these dependencies directly to 1NF, 2NF, and 3NF, explaining *why* we remove them

[Functional Dependency in DBMS: Cracking the Code - Simplilearn](#)